

TIC 77210120

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 1.534431 d
Epoch (t0) 3480.6325 BTJD
Duration 0.1700 d
Depth 4.621 % (46,205 ppm)
SNR 250.3

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

INDEPENDENT CHECKS (f3i gates)

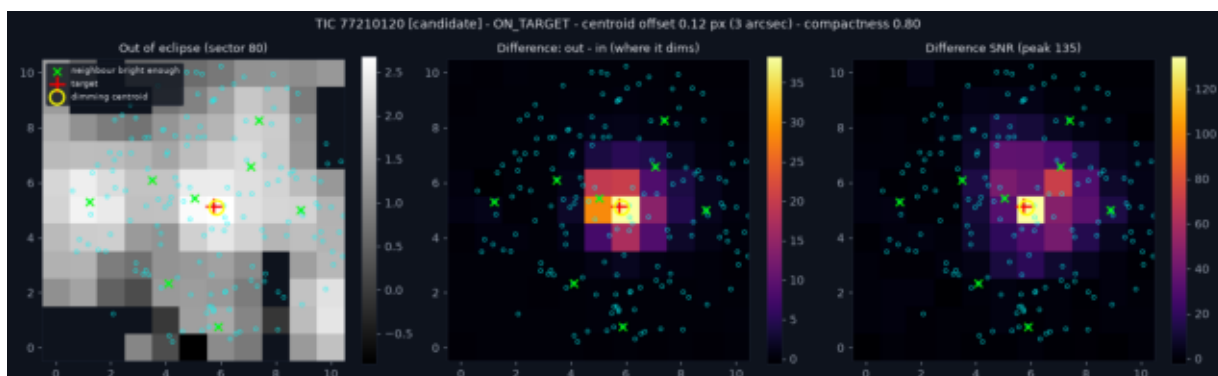
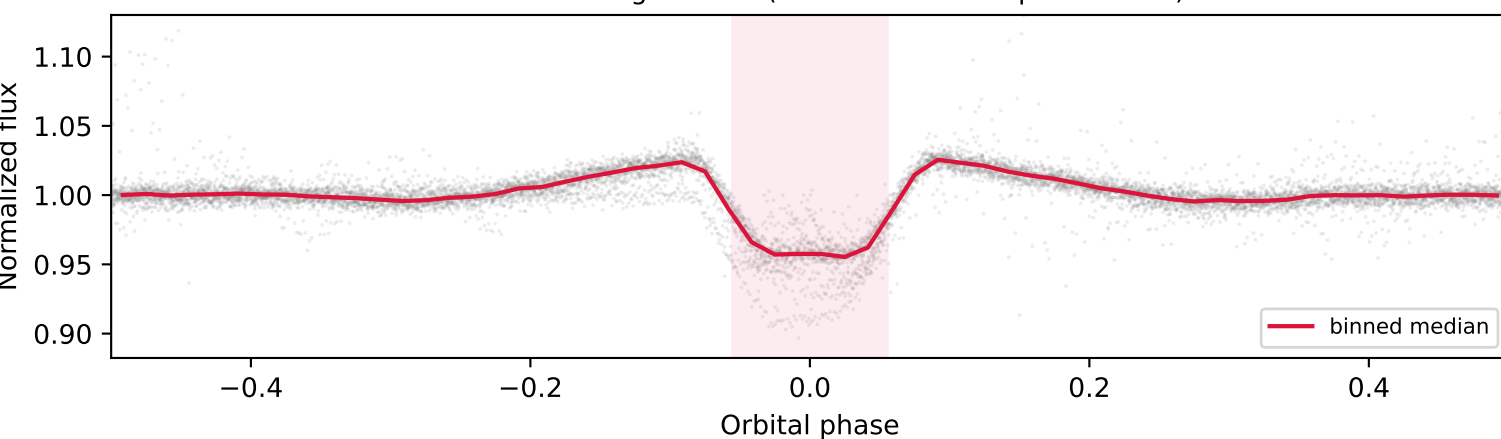
Eclipses observed 16 (gate ≥ 3)
Robust depth ratio 0.92 (gate ≥ 0.50)
Robust depth sigma 133.6 (gate ≥ 7.0)
Odd/even z-score 2.45 (consistent)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 80 light-curve dip in this sector: 123.5 sigma t0 refined by 0.0000 phase
Difference-image peak SNR 134.50 (gate ≥ 5.0) Compactness 0.80 (gate ≥ 0.50)
Dimming centroid offset 0.12 px (3 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 77210114
0.89 px away, dTmag +2.41 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.